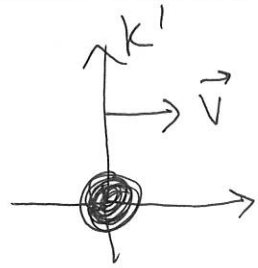
Transformation of angular distribution

<u>Application:</u>

piece of material with $\vec{v}$ emits gamma-rays (photons) uniformly in its rest frame K'.



What is the angular distribution of that radiation in K? (stationary lab frame)

in K':

$$\frac{dN'}{dt'\,d\Omega'} = \frac{I_0}{4\pi}$$

dN' = number of emitted particles during interval dt' into spherical angle $d\Omega'$

$d\Omega' = d\varphi'\, d\theta' \sin\theta'$.

in K: $dN = dN'$!

$dt = \gamma\, dt'$

$\varphi = \varphi'$ because it is in the plane perpendicular to $\vec{v}$.

Solid (spherical) angle $d\Omega'$ containing $(\theta', \theta'+d\theta')$ is then $d\Omega' = 2\pi \sin\theta'\, d\theta' = -2\pi\, d(\cos\theta')$

$$d\Omega = -2\pi\, d(\cos\theta)$$

Gamma-rays propagate with c (photons!)

$$\Rightarrow \quad v_x = c\cos\theta, \quad v_x' = c\cos\theta'$$

but $v_x' = \dfrac{v_x - v}{1 - \frac{v_x v}{c^2}} \Rightarrow c\cos\theta' = \dfrac{c\cos\theta - v}{1 - \frac{\cos\theta\, v}{c}}$

$$\cos\theta' = \frac{\cos\theta - \beta}{1 - \beta\cos\theta}$$

Then $d\Omega' = -2\pi\, d(\cos\theta') = -2\pi\, d\left(\frac{\cos\theta - \beta}{1-\beta\cos\theta}\right) =$

$$= -2\pi\left(\frac{-\sin\theta\, d\theta}{1-\beta\cos\theta} - \frac{\cos\theta-\beta}{(1-\beta\cos\theta)^2}\cdot\beta\sin\theta\, d\theta\right)$$

$$= \frac{2\pi\sin\theta\, d\theta}{1-\beta\cos\theta}\left(1 + \frac{\beta(\cos\theta-\beta)}{1-\beta\cos\theta}\right) =$$

$$= \frac{1-\beta^2}{(1-\beta\cos\theta)^2}\underbrace{d\Omega}_{2\pi\sin\theta\, d\theta}$$

$$\Rightarrow \frac{dN}{dt\, d\Omega} = \frac{dN'}{\gamma\, dt'\, d\Omega'}\frac{1-\beta^2}{(1-\beta\cos\theta)^2} = \frac{I_0}{4\pi}\frac{(1-\beta^2)^{3/2}}{(1-\beta\cos\theta)^2}$$

The distribution is anisotropic in K frame!

The emission is enhanced by $\frac{(1-\beta^2)^{3/2}}{(1-\beta)^2}$ over $\frac{I_0}{4\pi}$

for $\theta = 0$: $\frac{(1-\beta^2)^{3/2}}{(1-\beta)^2} = \frac{(1-\beta)^{3/2}(1+\beta)^{3/2}}{(1-\beta)^2} = \frac{(1+\beta)^{3/2}}{(1-\beta)^{1/2}} > 1$

It is suppresed by $(1-\beta^2)^{3/2} < 1$

for $\theta = \pi/2$.

This becomes particularly clear in the ultra-relativistic limit $\beta = 1-\eta$, with $\eta \ll 1$:

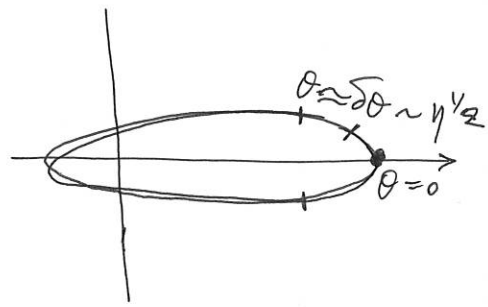
$$1-\beta^2 = (1+\beta)(1-\beta) \simeq 2\eta\ ,\quad 1-\beta\cos\theta \cong 1-(1-\eta)(1-\tfrac{1}{2}\theta^2) =$$

$$= 1-\left(1-\eta-\tfrac{1}{2}\theta^2+\cancel{\tfrac{\eta\theta^2}{2}}+\dots\right)$$

$$= \eta + \tfrac{1}{2}\theta^2$$

Then $\dfrac{dN}{dt\,d\Omega} \to \dfrac{I_0}{4\pi} \dfrac{(2\eta)^{3/2}}{\left(\eta + \frac{\theta^2}{2}\right)^2}$

Distribution is very narrow — has width $\delta\theta \sim \eta^{1/2} \ll 1$



Check the total number of particles emitted per unit time:

$$\frac{dN}{dt} = \int d\Omega \frac{dN}{dt\,d\Omega} = 2\pi \frac{I_0}{4\pi} \int_0^{\pi} d\theta \sin\theta \frac{(1-\beta^2)^{3/2}}{(1-\beta\cos\theta)^2}$$

$$= \frac{I_0 (1-\beta^2)^{3/2}}{2} \int_{-1}^{1} d(\cos\theta) \frac{1}{(1-\beta\cos\theta)^2} =$$

$$= \frac{I_0 (1-\beta^2)^{3/2}}{2\beta} \underbrace{\frac{+1}{1-\beta\cos\theta}\Bigg|_{-1}^{1}} = \frac{1}{2\beta} I_0 (1-\beta^2)^{3/2} \left(\frac{1}{1-\beta} - \frac{1}{1+\beta}\right)$$

$$\Rightarrow \frac{dN}{dt} = \frac{I_0 (1-\beta^2)^{3/2}}{2\beta} \frac{2\beta}{1-\beta^2} = I_0 (1-\beta^2)^{1/2} = \frac{I_0}{\gamma}.$$

As it should be because $\dfrac{dN'}{dt'} = I_0$ and $dt = \gamma\, dt'$.

$$\frac{dN}{dt} = \frac{dN'}{\gamma\, dt'} = \frac{I_0}{\gamma}.$$